MiniMotors

Business Description

Company Name: Mini Motors
Industry: Online Toy Retail & Manufacturing

Overview:
Mini Motors is a global online retailer specializing in transportation-themed toys such as cars, planes, and ships. The company manufactures toys in its factories and sells them to both individual customers and small toy businesses worldwide. Over the past two years (2023–2024), the company has experienced rapid growth, leading to an increase in sales volume and operational complexity.

Business Process & Challenges:
As sales expanded, managing orders and financial performance became increasingly difficult. The company faced several key challenges:

- Lack of clear insights into best-selling products across different regions and customer segments.
- Difficulty in tracking revenue, profitability, and sales trends over time.
- Inefficient identification of loyal customers who should receive targeted promotions.
- Limited visibility into the performance of business vs. individual customers.

These issues arose due to fragmented and unstructured sales data, making it challenging to make data-driven decisions.

Purpose of the Report:
This report is designed to provide Mini Motors with a structured analytical view of its sales performance using a star schema model. The report aims to:

- Identify high-performing products and regions based on sales trends.
- Evaluate revenue streams and profitability across different customer types.
- Optimize marketing efforts by identifying loyal and high-value customers.
- Improve decision-making through better data visualization and analysis.

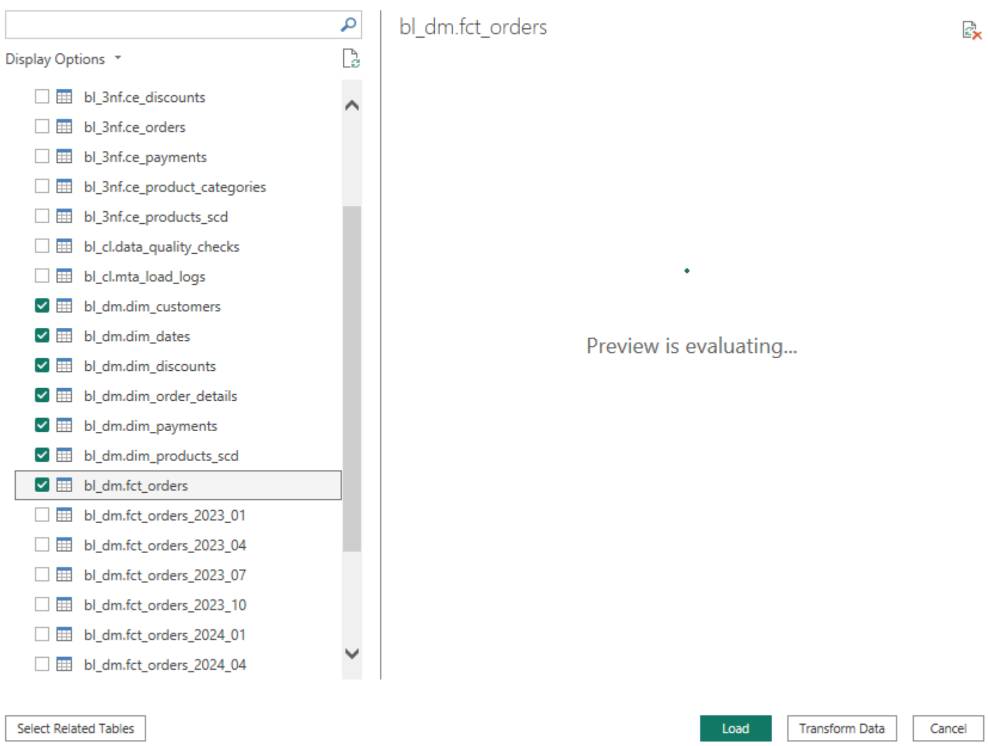
Target Audience:
The report is intended for company executives, sales managers, and financial analysts who

need actionable insights to optimize sales strategies, improve customer engagement, and enhance financial planning.

By addressing these challenges with a data-driven approach, Mini Motors can streamline operations, increase profitability, and maintain its competitive position in the global toy market.

Data import from PostgreSQL to POWER BI

Before reporting and visualizations, connection was established with PostgreSQL database via import mode where localhost:5432 was used as a server name which is a default port for PostgreSQL and MiniMotors as a database name. As a result, all tables from that database became visible in Power BI as well and all necessary entities were selected from bl_dm schema, which is a star model:



The data was already clean and well organized, and no additional transformations were needed so the use of Power Query was not necessary.

One of the most important details is checking the model, relationships and cardinality, everything was all right after importing data and here as well, no activation/deactivation or changes were needed:

Here are the relationships:

New relationship | Autodetect | Edit | Delete | Filter

From: table (column)	Relationship	To: table (column)	Status
bl_dm fct_orders (cust_surr_id)	* — 1	bl_dm dim_customers (cust_su...	Active
bl_dm fct_orders (discount_sur...	* — 1	bl_dm dim_discounts (discoun...	Active
bl_dm fct_orders (order_dt)	* — 1	bl_dm dim_dates (order_dt)	Active
bl_dm fct_orders (order_surr_id)	1 — 1	bl_dm dim_order_details (orde...	Active
bl_dm fct_orders (payment_sur...	1 — 1	bl_dm dim_payments (paymen...	Active
bl_dm fct_orders (prod_surr_id)	* — 1	bl_dm dim_products_scd (pro...	Active

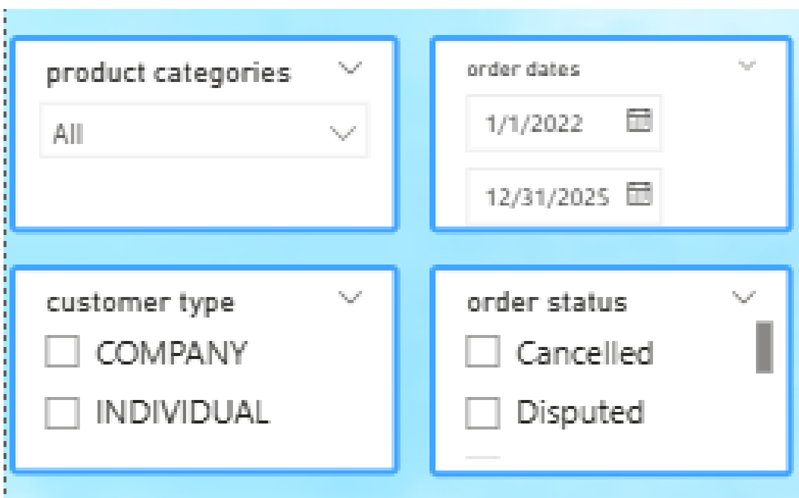
Reporting and Description of Visuals

The first page of the report

Filters

- **Product Categories**: Allows users to filter data based on specific toy categories (e.g., Cars, Planes, Ships).
- **Order Dates**: Users can set a date range (from January 1, 2022, to December 31, 2025) to analyze sales trends over time.
- **Customer Type**: Users can filter sales between COMPANY and INDIVIDUAL
- **Order Status**: can be used to filter out Cancelled and Disputed orders, or check only shipped orders ensuring only valid transactions are analyzed.

Purpose: These filters provide flexibility in data exploration, making the report interactive. End users can refine their view based on specific requirements.

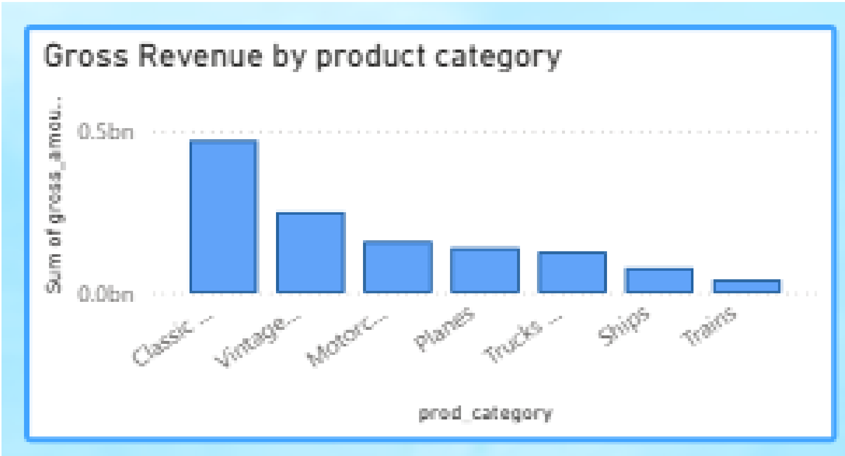


1. Gross Revenue by Product Category

Chart Type: Bar Chart

- Displays total gross revenue (before discounts) for each product category.
- Categories such as Classic, Vintage, and Motorcycle toys generate the highest revenue.

Purpose: Identifies best-performing product categories and their contribution to overall revenue.

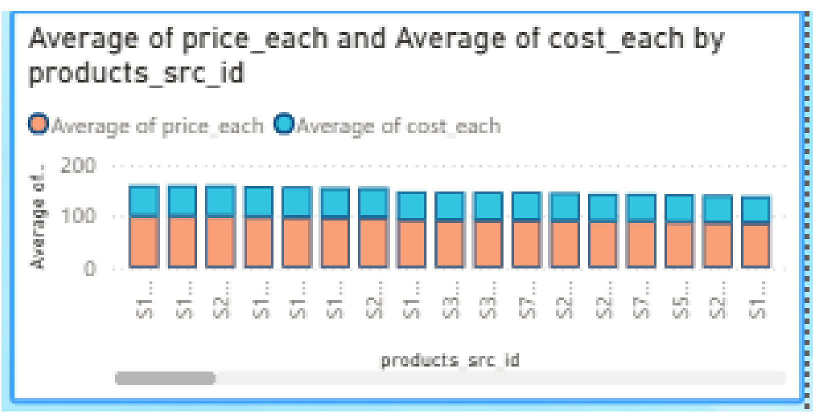


2. Average Price Each & Average Cost Each by Product

Chart Type: Clustered Bar Chart

- Compares the average selling price per unit (blue bars) and the average cost per unit (orange bars) for different products.
- A significant difference between price and cost indicates high-profit margins, whereas a small difference suggests lower profitability.

Purpose: Provides insight into product-level profitability and helps determine whether pricing adjustments are necessary.

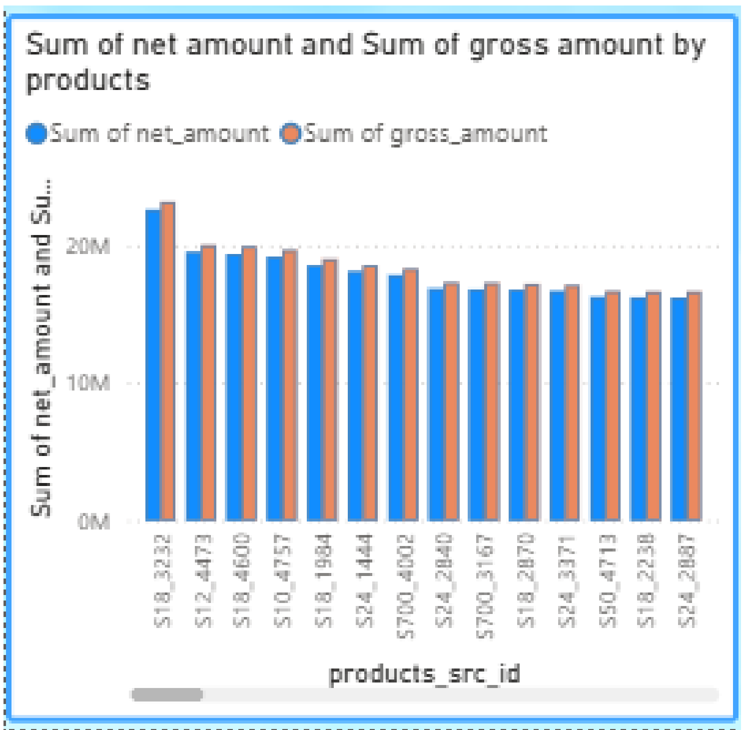


3. Sum of Net Amount & Sum of Gross Amount by Products

Chart Type: Bar Chart

- Displays net revenue (after discounts) versus gross revenue (before discounts) for each product.
- A significant difference between net and gross revenue suggests high discounts or customer returns.

Purpose: Evaluates the impact of discounts on revenue and identifies products with excessive discounts.

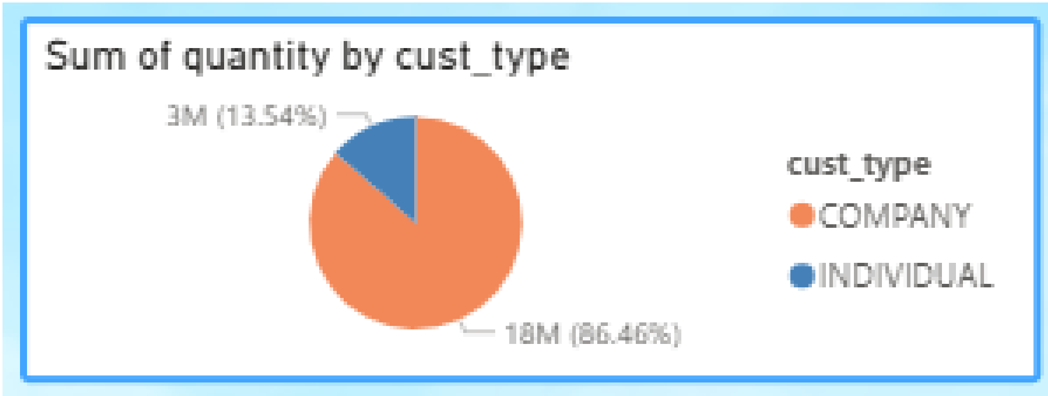


4. Sum of Quantity by Customer Type

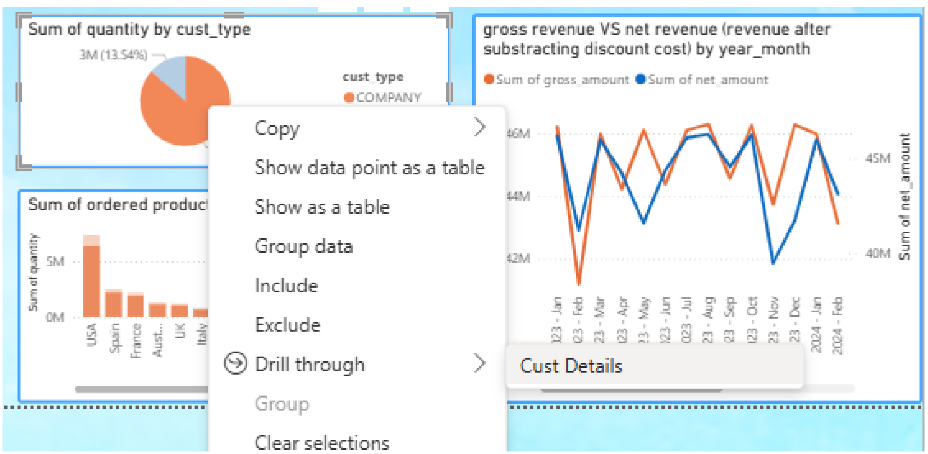
Chart Type: Pie Chart

- Compares total units sold between Companies and Individuals.
- In this dataset, 86.46% of sales come from companies, while 13.54% come from individual customers.

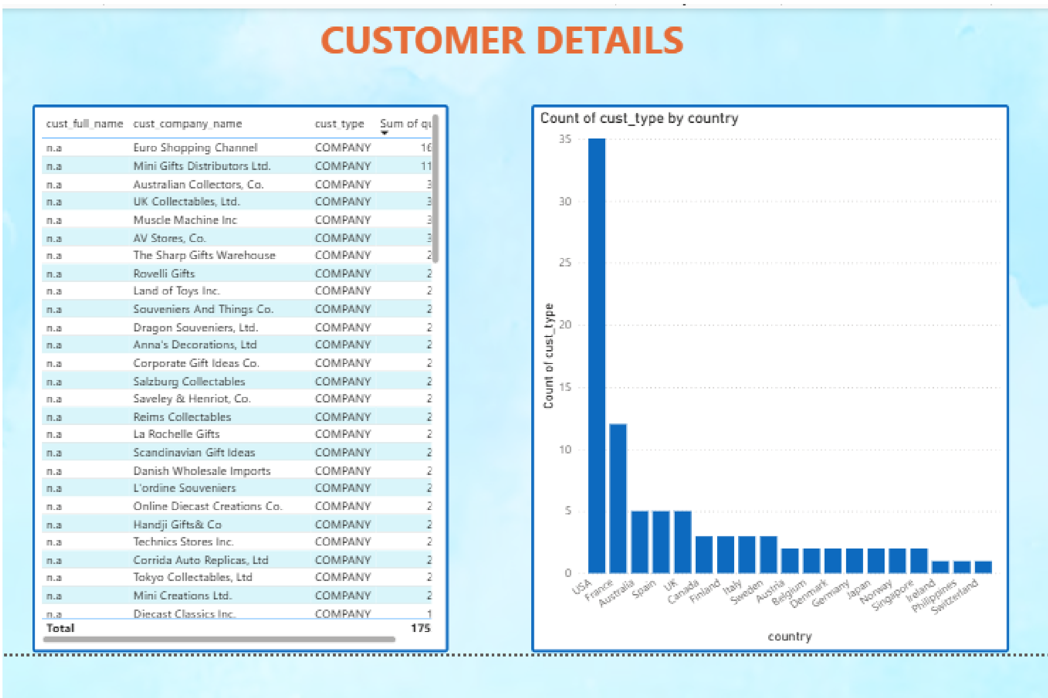
Purpose: Supports customer segmentation analysis and helps in developing targeted strategies for B2B vs. B2C markets.



Drillthrough is possible from this visual:



This action takes users to a whole new page where customer details are explained in more details. It is not necessary to have every single customers' detailed info and their sales amount displayed on charts and this feature allows users to inspect additional information in case of need and questions:



For this page specifically, new column cust_full_nmae was generated, which combines first name and last name while also considering default values:

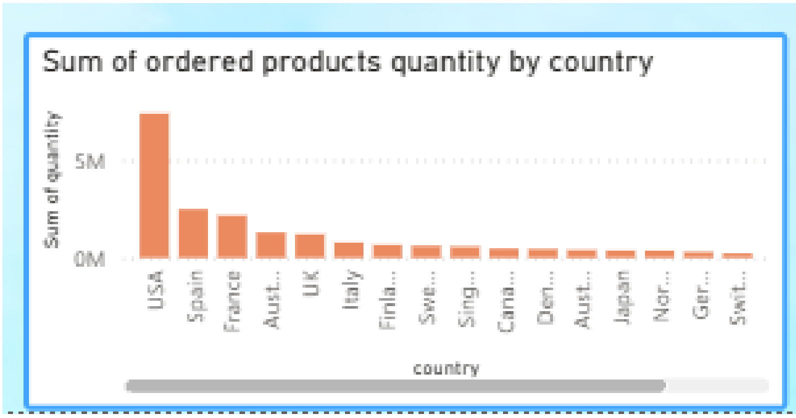
```
cust_full_name =
IF(
    OR(
        'bl_dm dim_customers'[cust_surr_id] = -1,
        AND('bl_dm dim_customers'[cust_first_name] = "n.a", 'bl_dm dim_customers'[cust_last_name] = "n.a")
    ),
    "n.a",
    COMBINEVALUES(" ",
        IF('bl_dm dim_customers'[cust_first_name] = "n.a", "", 'bl_dm dim_customers'[cust_first_name]),
        IF('bl_dm dim_customers'[cust_last_name] = "n.a", "", 'bl_dm dim_customers'[cust_last_name])
    )
)
```

5. Sum of Ordered Products Quantity by Country

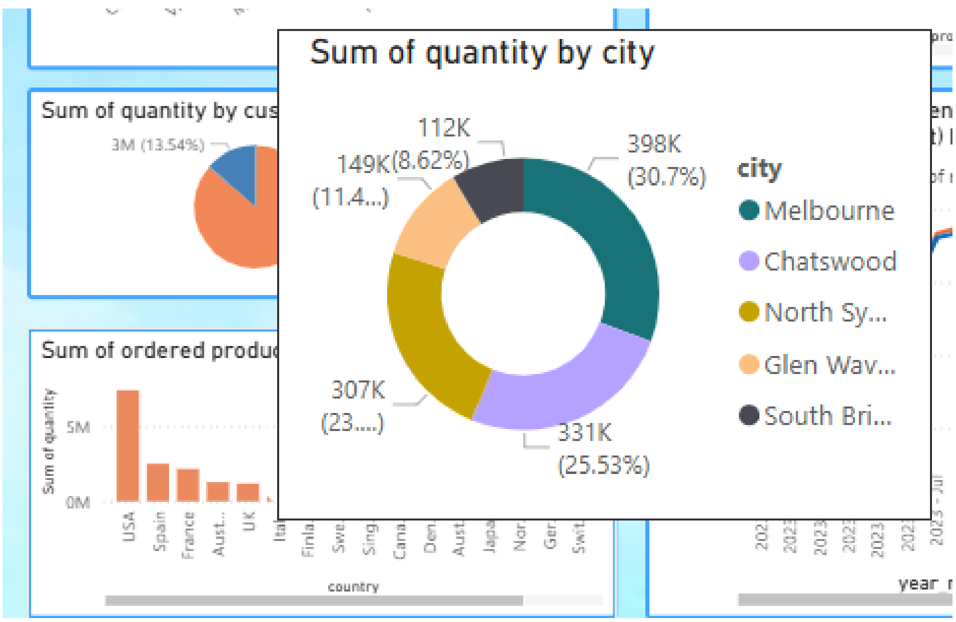
Chart Type: Bar Chart

- Displays the total quantity of products sold in each country.
- The USA has the highest number of orders, followed by Spain and France.

Purpose: Identifies key markets for expansion and prioritizes high-demand regions.



Tooltip was added for this chart as well, which additionally shows the cities' shares from each countries ordered quantity:

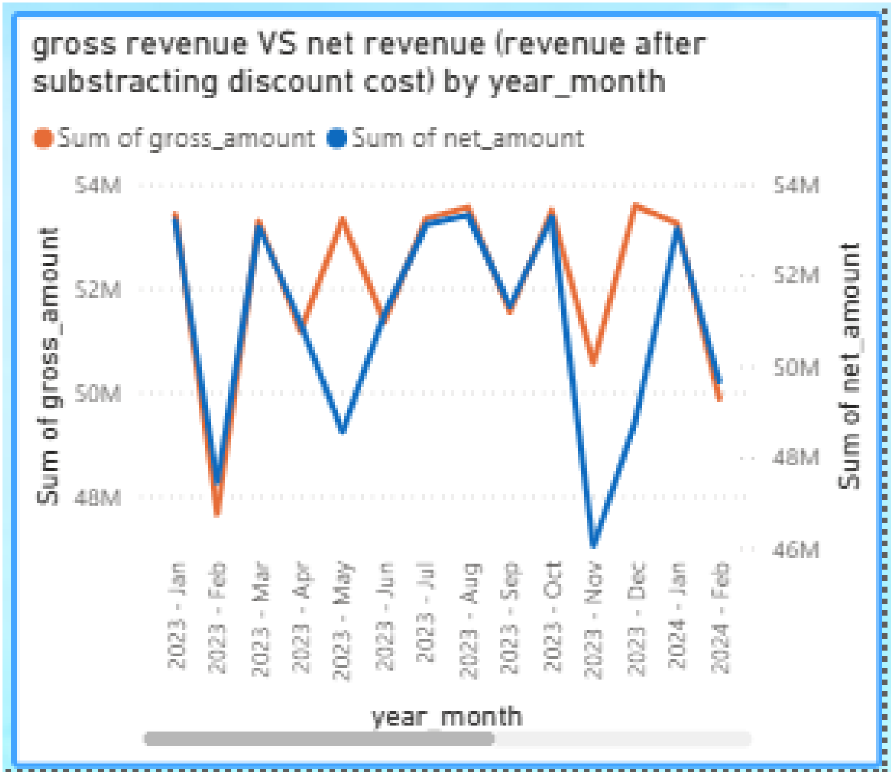


6. Gross Revenue vs. Net Revenue by Month

Chart Type: Line Chart

- The orange line represents Gross Revenue (before discounts).
- The blue line represents Net Revenue (after discounts).
- A significant gap between these lines indicates a high discount rate or substantial customer returns.

Purpose: Monitors monthly revenue trends, identifies seasonal patterns, and assesses profitability fluctuations.



Overall Importance of This Page

- Identifies best-selling product categories and their contribution to revenue.
- Compares profit margins using price and cost data.
- Tracks revenue performance across different months.
- Segments customers into B2B and B2C for targeted marketing.
- Highlights top-performing countries in terms of sales volume.

Business Implications:

- Optimizing pricing and discount strategies.
- Prioritizing high-revenue product categories.
- Improving inventory and supply chain planning.

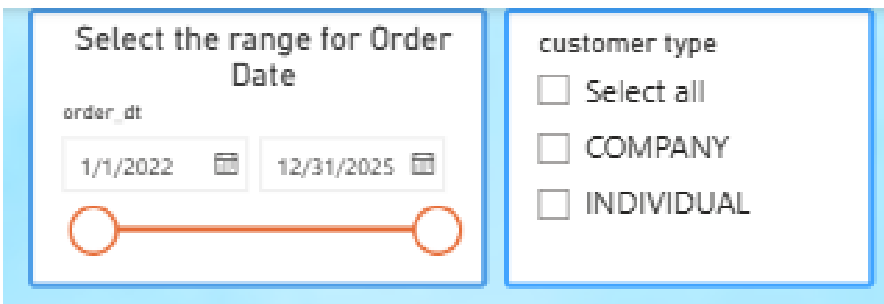
- Strengthening market presence in key regions.

Second page of the report

Filters

- **Order Dates:** Users can set a date range from January 1, 2022, to December 31, 2025 to analyze trends over time.
- **Customer Type:** Filters between COMPANY (B2B) and INDIVIDUAL (B2C) sales.
- **Select Time Period Range:** Allows users to choose between viewing data by order date, year-month, or year-quarter.

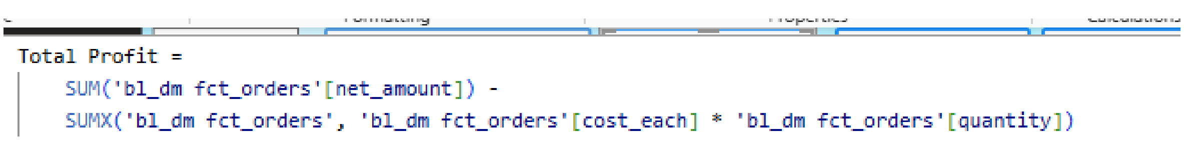
Purpose: These filters help users interactively refine their analysis by focusing on specific time periods, customer types, and valid transactions.



1. Total Profit by Product Category

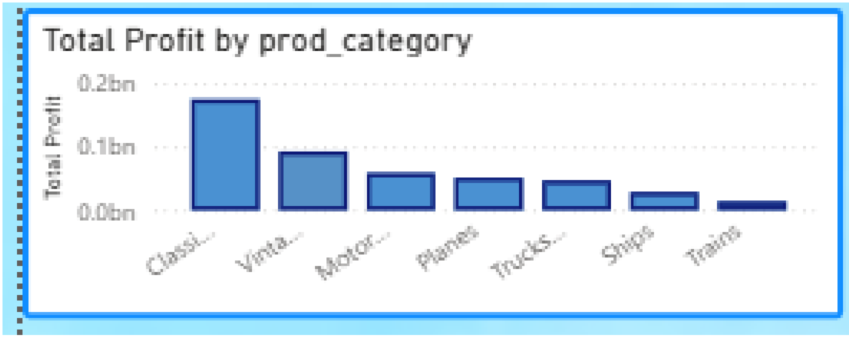
Chart Type: Bar Chart

- Shows total profit per product category.
- Categories like Classic, Motorcycles, and Vintage generate the highest profits.
- To achieve this, new measure "Total Profit" was created using Dax, which gets calculated by subtracting the cost of producing of product:



```
Total Profit =
    SUM('bl_dm fct_orders'[net_amount]) -
    SUMX('bl_dm fct_orders', 'bl_dm fct_orders'[cost_each] * 'bl_dm fct_orders'[quantity])
```

Purpose: Helps identify the most profitable product categories and guides business strategy for product prioritization.



2. Total Profit by year-month , Order_dt , year_quarter or year

Chart Type: Line Chart

- Tracks monthly profit trends from 2023 to 2024.
- Profits are analyzed based on the time period range chosen by the user. This is achieved using some calculated columns and parameter:

  Created calculated column year-month:

```
1 year_month = 'bl_dm dim_dates'[year] & " - " & left('bl_dm dim_dates'[month_name], 3)
```

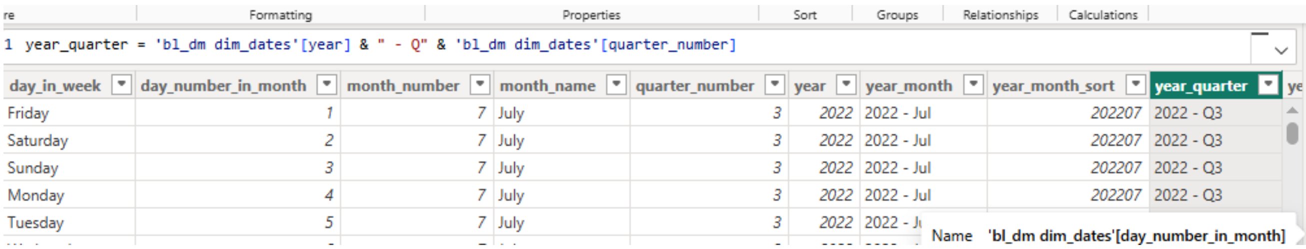
order_dt	day_in_week	day_number_in_month	month_number	month_name	quarter_number	year	year_month	year_month_sort	year_quar
Friday, July 1, 2022	Friday	1	7	July	3	2022	2022 - Jul	202207	2022 - (
Saturday, July 2, 2022	Saturday	2	7	July	3	2022	2022 - Jul	202207	2022 - (
Sunday, July 3, 2022	Sunday	3	7	July	3	2022	2022 - Jul	202207	2022 - (

  To sort this column correctly, I created this column as well:

```
1 year_month_sort = 'bl_dm dim_dates'[year] * 100 + 'bl_dm dim_dates'[month_number]
```

order_dt	day_in_week	day_number_in_month	month_number	month_name	quarter_number	year	year_month	year_month_sort	year_quar
Friday, July 1, 2022	Friday	1	7	July	3	2022	2022 - Jul	202207	2022 - (
Saturday, July 2, 2022	Saturday	2	7	July	3	2022	2022 - Jul	202207	2022 - (
Sunday, July 3, 2022	Sunday	3	7	July	3	2022	2022 - Jul	202207	2022 - (
Monday, July 4, 2022	Monday	4	7	July	3	2022	2022 - Jul	202207	2022 - (
Tuesday, July 5, 2022	Tuesday	5	7	July	3	2022	2022 - Jul	202207	2022 - (
Wednesday, July 6, 2022	Wednesday	6	7	July	3	2022	2022 - Jul	202207	2022 - (

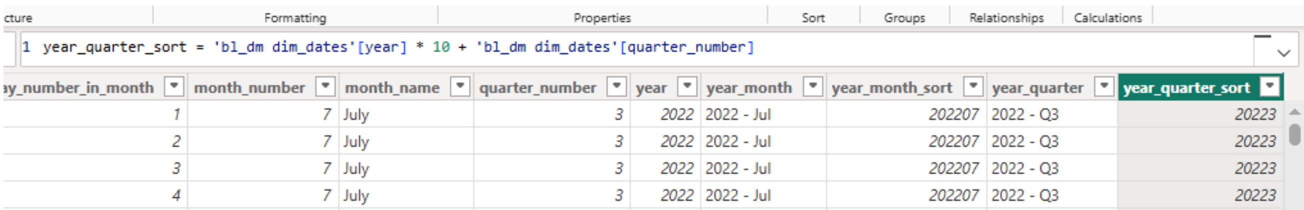
Created another calculated column year-quarter:



```
1 year_quarter = 'bl_dm dim_dates'[year] & " - Q" & 'bl_dm dim_dates'[quarter_number]
```

day_in_week	day_number_in_month	month_number	month_name	quarter_number	year	year_month	year_month_sort	year_quarter
Friday	1	7	July	3	2022	2022 - Jul	202207	2022 - Q3
Saturday	2	7	July	3	2022	2022 - Jul	202207	2022 - Q3
Sunday	3	7	July	3	2022	2022 - Jul	202207	2022 - Q3
Monday	4	7	July	3	2022	2022 - Jul	202207	2022 - Q3
Tuesday	5	7	July	3	2022	2022 - Ju		

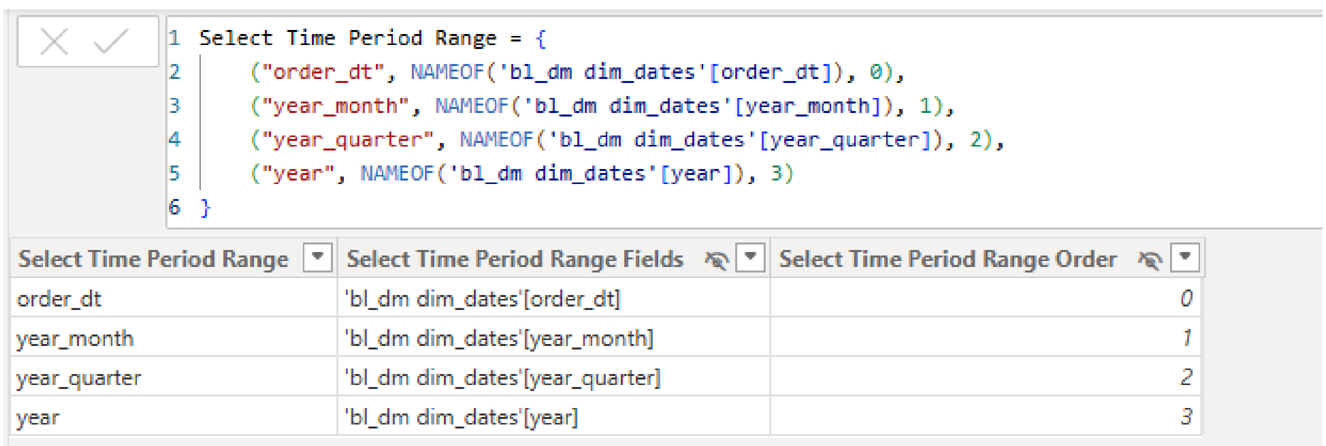
To sort this created one additional column:



```
1 year_quarter_sort = 'bl_dm dim_dates'[year] * 10 + 'bl_dm dim_dates'[quarter_number]
```

day_number_in_month	month_number	month_name	quarter_number	year	year_month	year_month_sort	year_quarter	year_quarter_sort
1	7	July	3	2022	2022 - Jul	202207	2022 - Q3	20223
2	7	July	3	2022	2022 - Jul	202207	2022 - Q3	20223
3	7	July	3	2022	2022 - Jul	202207	2022 - Q3	20223
4	7	July	3	2022	2022 - Jul	202207	2022 - Q3	20223

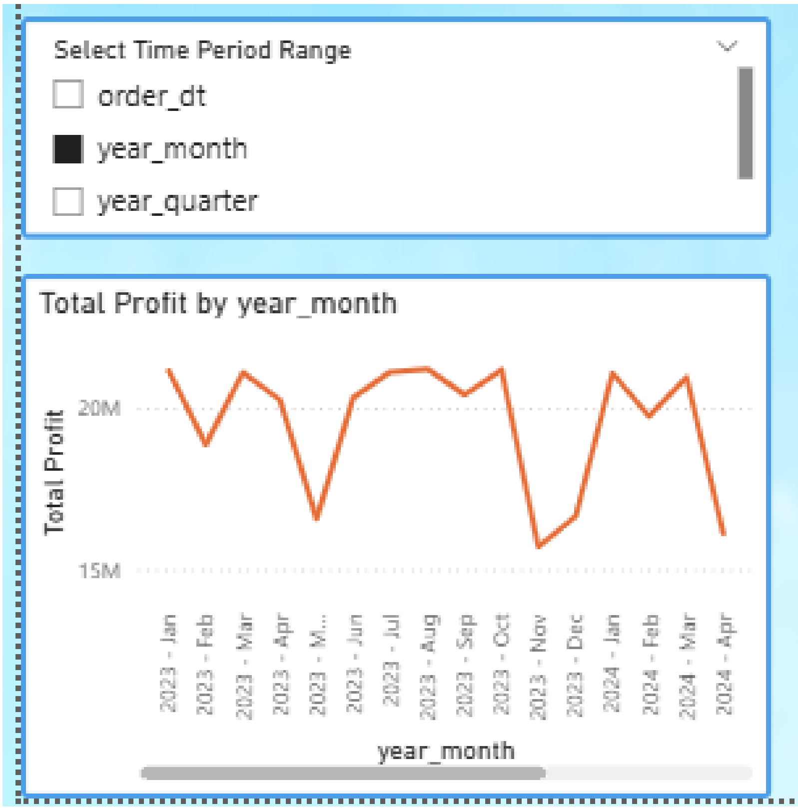
Then parameter was created:



```
Select Time Period Range = {
    ("order_dt", NAMEOF('bl_dm dim_dates'[order_dt]), 0),
    ("year_month", NAMEOF('bl_dm dim_dates'[year_month]), 1),
    ("year_quarter", NAMEOF('bl_dm dim_dates'[year_quarter]), 2),
    ("year", NAMEOF('bl_dm dim_dates'[year]), 3)
}
```

Select Time Period Range	Select Time Period Range Fields	Select Time Period Range Order
order_dt	'bl_dm dim_dates'[order_dt]	0
year_month	'bl_dm dim_dates'[year_month]	1
year_quarter	'bl_dm dim_dates'[year_quarter]	2
year	'bl_dm dim_dates'[year]	3

After that parameter, the slicer was created automatically, which enables users to choose their desired range of time period:



- Shows fluctuations in profit over time, with peaks and drops.

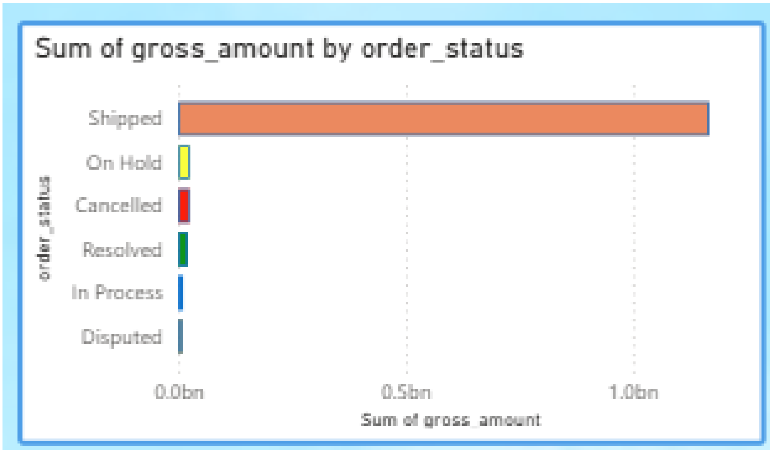
Purpose: Helps monitor long-term profitability trends, identify seasonality, and assess the impact of business decisions on revenue.

3. Sum of Gross Amount by Order Status

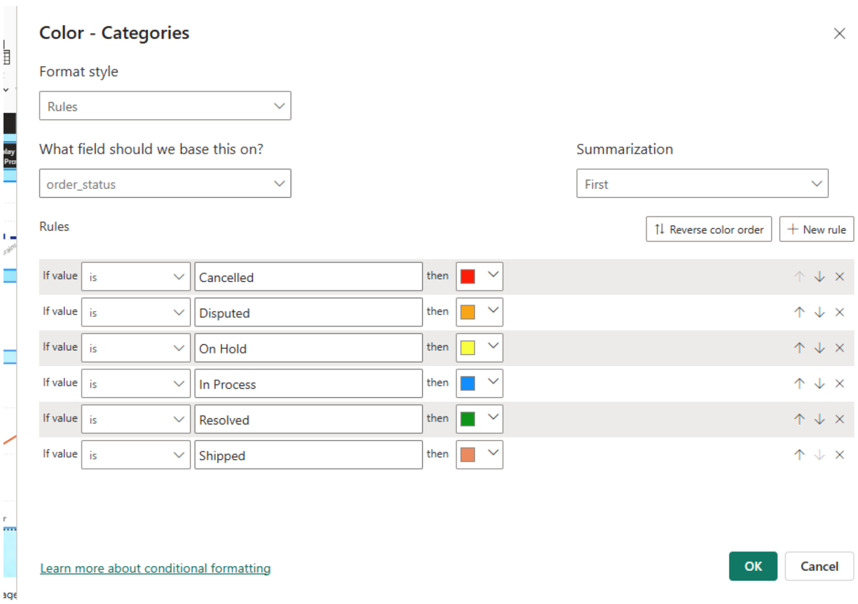
Chart Type: Horizontal Bar Chart

- Breaks down total gross revenue by order status.
- The majority of revenue comes from Shipped orders.
- Other statuses like On Hold, Resolved, In Process and others contribute minimally.

Purpose: Highlights the distribution of order statuses, helping in logistics and fulfillment optimization.



Conditional formatting was used to differentiate order statuses based on colors:

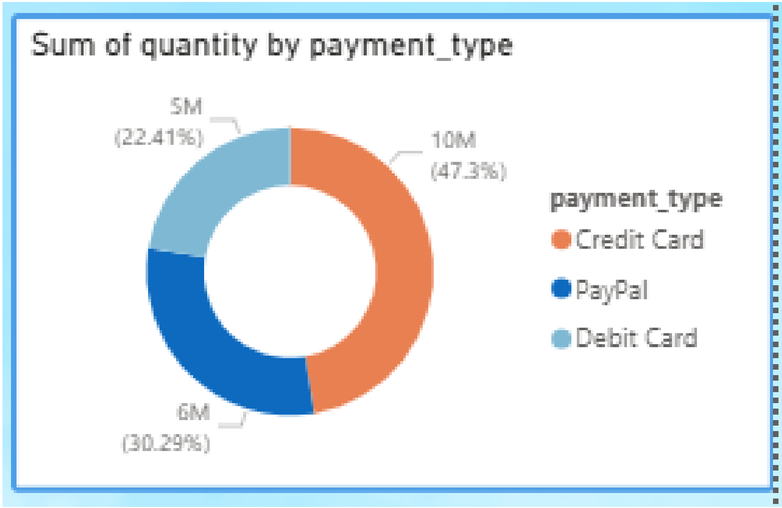


4. Sum of Quantity by Payment Type

Chart Type: Donut Chart

- Compares total quantity sold based on payment type.
- Credit Cards (47.3%) dominate transactions, followed by PayPal(30.29%) and DebitCard(22.41%).

Purpose: Helps understand customer payment preferences, enabling businesses to tailor payment options for convenience and higher sales.

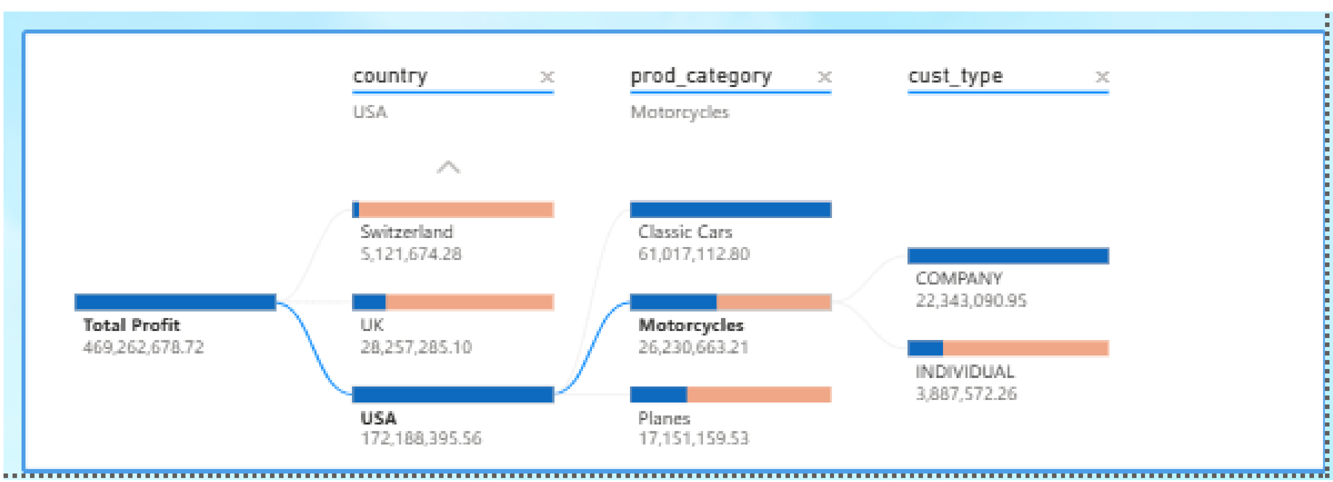


5. Profit Breakdown by Country, Product Category, and Customer

Chart Type: Decomposition Tree

- Visualizes profit distribution across countries, product categories, and customer types.
- Total profit is represented as a dependent variable and country , product category and customer_type is used as an independent variable.

Purpose: Offers a multi-dimensional view of where profits come from, aiding in geographic expansion, product focus, and B2B vs. B2C strategy planning.



6. Key Financial Metrics (Top Right Summary Cards)

- Total Profit: 469.26M
- Total Revenue: 1.25bn

Purpose: Provides a high-level overview of business performance, making it easy to track key financial indicators.

Total Profit measure already existed and now one additional measure Total revenue was generated as well:

```
Total Revenue = SUM('bl_dm fct_orders'[gross_amount])
```

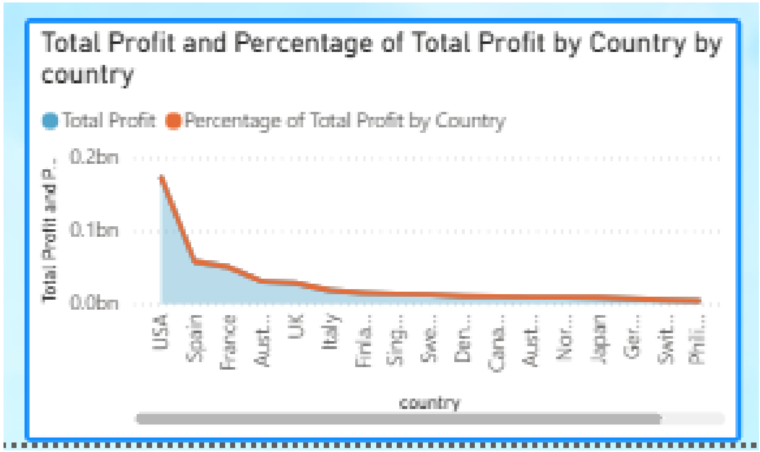
Total Profit and Percentage of Total Profit by Country
Chart Type: Stacked Area Chart

- Displays total profit by country over time.
- Shows the percentage contribution of each country to overall profit.

To display this percentage contribution, new measure was created:

```
Percentage of Total Profit by Country =
DIVIDE(
    [Total Profit],
    CALCULATE(
        [Total Profit],
        ALL('bl_dm dim_customers'[country])
    ),
    0
)
```

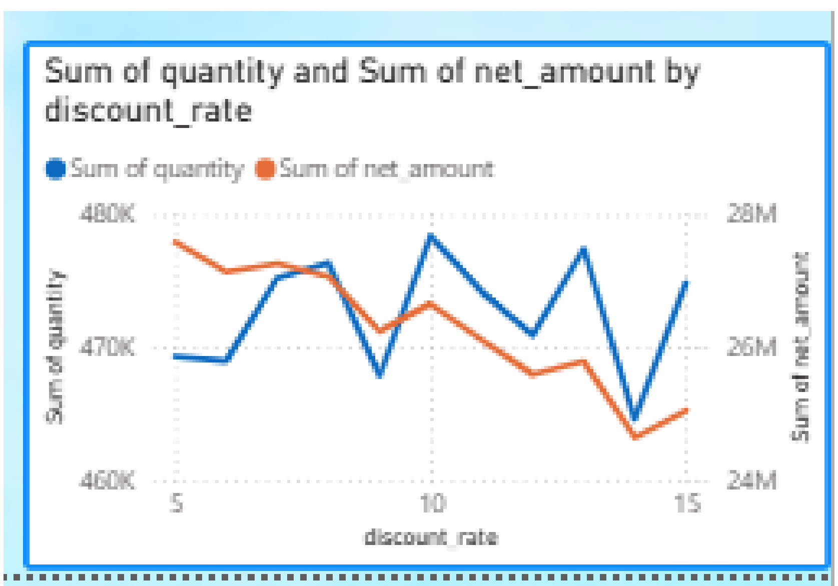
Purpose: Helps analyze geographic profit distribution, identify top-performing markets, and support expansion strategies.



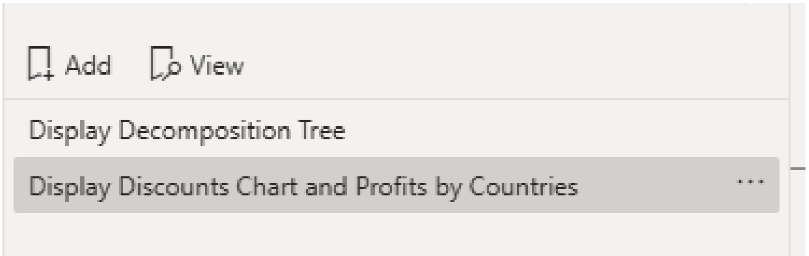
Sum of Quantity and Sum of Net Amount by Discount Rate
Chart Type: Stacked Column Chart

- Displays total quantity sold and net amount grouped by discount rate.

Purpose: Helps assess the impact of discount strategies on sales and revenue, guiding pricing and promotional decisions.



Bookmarks are used on this page:



And navigation through these two bookmarks are done using buttons:

This is the view with the second button on:

And this is with the first button:

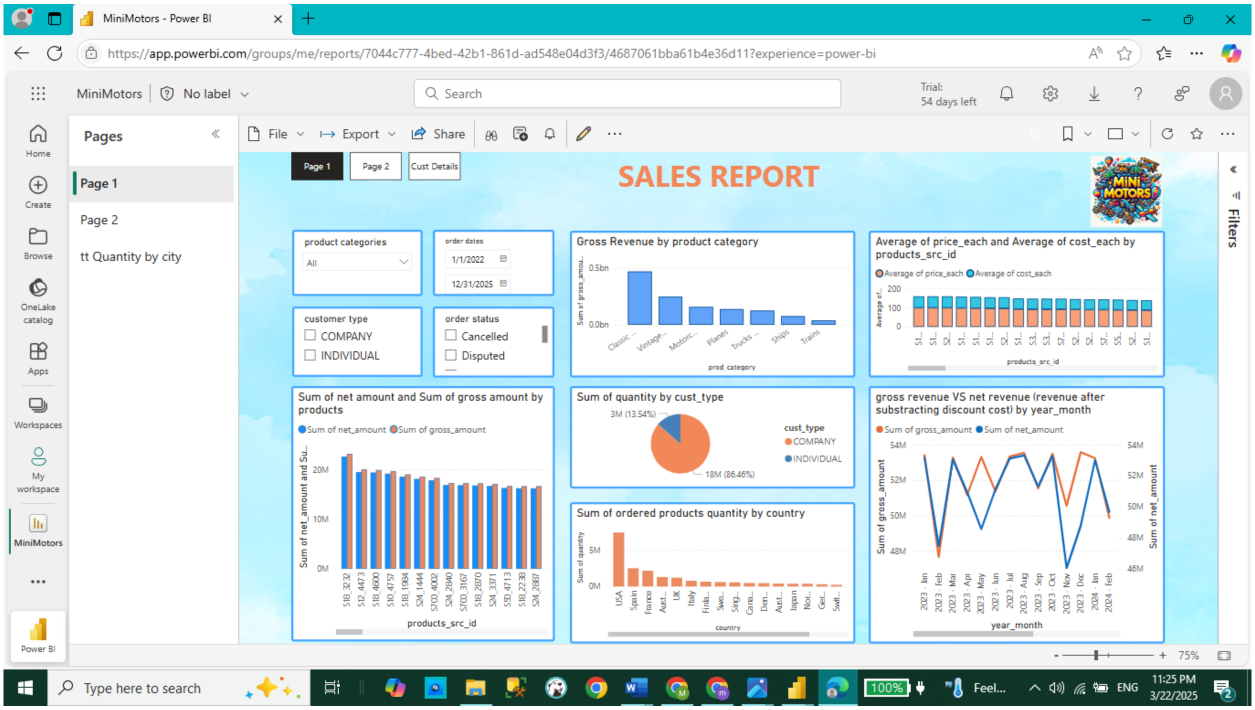
Overall Importance of This Page

- Tracks profitability across time periods, product categories, and customer segments.
- Analyzes revenue distribution based on order statuses and payment methods.
- Provides insights into geographical market performance and customer preferences.
- Helps refine sales strategies by identifying top-selling categories and high-profit regions.

Business Implications

- Optimize product focus by prioritizing profitable categories.
- Adjust marketing strategies based on payment preferences.
- Improve fulfillment processes by analyzing order status distribution.
- Expand in high-profit regions like the USA and UK.
- Tailor B2B and B2C strategies for maximum profitability.

Finally, this report was published in Power BI service:



And now end users: company executives, sales managers, and financial analysts can check it to gain insights.